SIMULTANEOUS LOCAL FACTORIZATION OF NONZERODIVISORS AFTER A FINITE EXTENSION OF A 1-DIMENSIONAL NOETHERIAN LOCAL RING

THE STACKS PROJECT AUTHORS

Editorial note. Counted proposition: Tag 0EAG, chow.tex lines 641-655 with its own proof at lines 657-752 (statement immediately followed by its proof; no non-adjacent pairing).

Editorial note. The four context blocks reproduce the author's own definitions and the three supporting lemmas of the same section (Tags 02PH, 0EA6, 0EA9, 0EAE, 02Q7, 0EAF). They are supporting results, not additional counted problems.

Editorial note. Every external reference is a navigation pointer to the stable Stacks tag; none of them replaces the counted construction or the length induction retained in this item.

Editorial note. The retained preamble is the project's native preamble (Xy-pic setup and the theorem/macro block), not a new macro layer; documentclass amsart is supplied mechanically because the native preamble selects the project class file, which is not redistributed here.

Allowed standard prerequisites (editorial).

- Formal glueing of finite modules and algebras over a ring with finitely many maximal ideals (More on Algebra, Lemma 0ALK; Remark 05EU). Stacks tags 0ALK, 05EU. Produces the finite extension B contained in C whose localizations realize prescribed local extensions; applied by the author at chow.tex lines 681-693 and quoted again at 549-558.
- Finiteness of the blow-up of a 1-dimensional local ring in a two-generator ideal (Divisors, Lemmas 02OS, 02NS, 0BFM; Varieties, Lemma 0AB7). Stacks tags 02OS, 02NS, 0BFM, 0AB7. Supplies R contained in R' with a = t a', b = t b' and R' = a'R' + b'R' in the second retained lemma (chow.tex lines 575-601).
- Domination of a 1-dimensional local domain by a discrete valuation ring, and pushdown of lengths along a finite local extension. Stacks tags 00P8, 02M0 (Algebra, Lemmas 00P8 and 02M0). Bounds the exponent m in the third retained lemma (chow.tex lines 612-638) and computes length_A(A'/xA') at chow.tex lines 716-722.
- Comparison of multiplicities of 2-periodic complexes under a finite-length kernel and cokernel. Stacks tags 02PH, 0EA6, 0EA9 (retained context, source lines 165-180, 197-206, 282-288). Gives length_A(A/xA) = length_A(A'/xA') in the author's proof at chow.tex lines 713-715.

Source reference. Original lemma 0ALK: sources/stacks-chow/tags.txt, source lines 13373–13373. Navigation pointer to the formal glueing lemma cited at chow.tex line 549; the counted content is the induction of lines 657-752 that applies it, not the glueing result itself.

Date: stacks-project:chow.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original remark 05EU: sources/stacks-chow/tags.txt, source lines 7003–7003. Remark cited at chow.tex line 553 for endowing the glued module with an algebra structure; the author's argument is unchanged and the pointer carries no proof obligation of this task.

Source reference. Original lemma 00P9: sources/stacks-chow/tags.txt, source lines 923–923. Quoted at chow.tex lines 562-563 only to say that the following supporting lemma is very similar to it; no step of the counted proof depends on its statement.

Source reference. Original lemma 02OS: sources/stacks-chow/tags.txt, source lines 3527–3527. Standard blow-up input cited at chow.tex lines 581-582 and 596 inside the retained supporting lemma; the pointer replaces neither that lemma nor the induction of the counted proof.

Source reference. Original lemma 02NS: sources/stacks-chow/tags.txt, source lines 3491–3491. Projectivity of the blow-up, cited at chow.tex line 587 as a standard scheme-theoretic fact inside the retained supporting lemma.

Source reference. Original lemma 0BFM: sources/stacks-chow/tags.txt, source lines 14390–14390. Used at chow.tex line 589 to see that every generic point of the blow-up lies over the punctured spectrum; a standard pointer inside retained context.

Source reference. Original lemma 0AB7: sources/stacks-chow/tags.txt, source lines 13010–13010. Finiteness criterion cited at chow.tex line 590 to conclude that the blow-up is affine; a standard pointer inside retained context.

Source reference. Original lemma 02M9: sources/stacks-chow/tags.txt, source lines 3436–3436. Cited at chow.tex line 599 to turn the locally principal ideal IR' into a principal one over a semi-local ring; standard commutative-algebra pointer.

Source reference. Original lemma 00P8: sources/stacks-chow/tags.txt, source lines 922–922. Existence of a dominating discrete valuation ring, cited twice at chow.tex lines 615-617 and 633-634 inside the retained boundedness lemma; the pointer does not replace the counted induction.

Source reference. Original lemma 00GQ: sources/stacks-chow/tags.txt, source lines 616–616. Surjectivity of Spec on integral extensions, cited at chow.tex line 626 inside retained context.

Source reference. Original lemma 00FI: sources/stacks-chow/tags.txt, source lines 572–572. Surjectivity of Spec after the radical base change A to A', cited at chow.tex line 628 inside retained context.

Source reference. Original lemma 02M0: sources/stacks-chow/tags.txt, source lines 3427–3427. Length pushdown along a finite extension, cited at chow.tex line 722 in the author's proof; it is an allowed foundational input and the length induction that uses it is retained in full.

Definition 0.1. Let (M, N, φ, ψ) be a 2-periodic complex over a ring R whose cohomology modules have finite length. In this case we define the *multiplicity* of (M, N, φ, ψ) to be the integer

$$e_R(M, N, \varphi, \psi) = \text{length}_R(H^0(M, N, \varphi, \psi)) - \text{length}_R(H^1(M, N, \varphi, \psi))$$

In the case of a $(2, 1)$-periodic complex (M, φ, ψ), we denote this by $e_R(M, \varphi, \psi)$ and we will sometimes call this the *(additive) Herbrand quotient.*

An example of a $(2,1)$-periodic complex over a ring R is any triple of the form $(M,0,\psi)$ where M is an R-module and ψ is an R-linear map. If the kernel and cokernel of ψ have finite length, then we obtain

$$e_R(M,0,\psi) = \text{length}_R(\text{Coker}(\psi)) - \text{length}_R(\text{Ker}(\psi)) \tag{0.1.1}$$

We state and prove the obligatory lemmas on these notations.

Lemma 0.2. *Let R be a ring. Let $f : (M,\varphi,\psi) \to (M',\varphi',\psi')$ be a map of $(2,1)$-periodic complexes whose cohomology modules have finite length. If $\text{Ker}(f)$ and $\text{Coker}(f)$ have finite length, then $e_R(M,\varphi,\psi) = e_R(M',\varphi',\psi')$.*

1. Preparation for tame symbols

In this section we put some lemma that will help us define the tame symbol in the next section.

Lemma 1.1. *Let A be a Noetherian ring. Let $\mathfrak{m}_1,\ldots,\mathfrak{m}_r$ be pairwise distinct maximal ideals of A. For $i = 1,\ldots,r$ let $\varphi_i : A_{\mathfrak{m}_i} \to B_i$ be a ring map whose kernel and cokernel are annihilated by a power of $\mathfrak{m}_i$. Then there exists a ring map $\varphi : A \to B$ such that*

1. *the localization of φ at $\mathfrak{m}_i$ is isomorphic to φ_i, and*
2. *$\text{Ker}(\varphi)$ and $\text{Coker}(\varphi)$ are annihilated by a power of $\mathfrak{m}_1 \cap \ldots \cap \mathfrak{m}_r$.*

Moreover, if each φ_i is finite, injective, or surjective then so is φ.

Proof. Set $I = \mathfrak{m}_1 \cap \ldots \cap \mathfrak{m}_r$. Set $A_i = A_{\mathfrak{m}_i}$ and $A' = \prod A_i$. Then $IA' = \prod \mathfrak{m}_i A_i$ and $A \to A'$ is a flat ring map such that $A/I \cong A'/IA'$. Thus we may use More on Algebra, Lemma 0ALK to see that there exists an A-module map $\varphi : A \to B$ with φ_i isomorphic to the localization of φ at $\mathfrak{m}_i$. Then we can use the discussion in More on Algebra, Remark 05EU to endow B with an A-algebra structure matching the given A-algebra structure on B_i. The final statement of the lemma follows easily from the fact that $\text{Ker}(\varphi)_{\mathfrak{m}_i} \cong \text{Ker}(\varphi_i)$ and $\text{Coker}(\varphi)_{\mathfrak{m}_i} \cong \text{Coker}(\varphi_i)$. □

The following lemma is very similar to Algebra, Lemma 00P9.

Lemma 1.2. *Let $(R,\mathfrak{m})$ be a Noetherian local ring of dimension 1. Let $a,b \in R$ be nonzerodivisors. There exists a finite ring extension $R \subset R'$ with R'/R annihilated by a power of $\mathfrak{m}$ and nonzerodivisors $t,a',b' \in R'$ such that $a = ta'$ and $b = tb'$ and $R' = a'R' + b'R'$.*

Proof. If a or b is a unit, then the lemma is true with $R = R'$. Thus we may assume $a,b \in \mathfrak{m}$. Set $I = (a,b)$. The idea is to blow up R in I. Instead of doing the algebraic argument we work geometrically. Let $X = \text{Proj}(\bigoplus_{d\geq 0} I^d)$. By Divisors, Lemma 02OS the morphism $X \to \text{Spec}(R)$ is an isomorphism over the punctured spectrum $U = \text{Spec}(R) \setminus \{\mathfrak{m}\}$. Thus we may and do view U as an open subscheme of X. The morphism $X \to \text{Spec}(R)$ is projective by Divisors, Lemma 02NS. Also, every generic point of X lies in U, for example by Divisors, Lemma 0BFM. It follows from Varieties, Lemma 0AB7 that $X \to \text{Spec}(R)$ is finite. Thus $X = \text{Spec}(R')$ is affine and $R \to R'$ is finite. We have $R_a \cong R'_a$ as $U = D(a)$. Hence a power of a annihilates the finite R-module R'/R. As $\mathfrak{m} = \sqrt{(a)}$ we see that R'/R is annihilated by a power of $\mathfrak{m}$. By Divisors, Lemma 02OS we see that IR' is a locally principal ideal. Since R' is semi-local we see that IR' is principal, see Algebra, Lemma 02M9, say $IR' = (t)$. Then we have $a = a't$ and $b = b't$ and everything is clear. □

Lemma 1.3. *Let $(R,\mathfrak{m})$ be a Noetherian local ring of dimension 1. Let $a,b \in R$ be nonzerodivisors with $a \in \mathfrak{m}$. There exists an integer $n = n(R,a,b)$ such that for a finite ring extension $R \subset R'$ if $b = a^m c$ for some $c \in R'$, then $m \leq n$.*

Proof. Choose a minimal prime $\mathfrak{q} \subset R$. Observe that $\dim(R/\mathfrak{q}) = 1$, in particular $R/\mathfrak{q}$ is not a field. We can choose a discrete valuation ring A dominating $R/\mathfrak{q}$ with the same fraction field, see Algebra, Lemma 00P8. Observe that a and b map to nonzero elements of A as nonzerodivisors in R are not contained in $\mathfrak{q}$. Let v be the discrete valuation on A. Then $v(a) > 0$ as $a \in \mathfrak{m}$. We claim $n = v(b)/v(a)$ works.

Let $R \subset R'$ be given. Set $A' = A \otimes_R R'$. Since $\mathrm{Spec}(R') \to \mathrm{Spec}(R)$ is surjective (Algebra, Lemma 00GQ) also $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is surjective (Algebra, Lemma 00FI). Pick a prime $\mathfrak{q}' \subset A'$ lying over $(0) \subset A$. Then $A \subset A'' = A'/\mathfrak{q}'$ is a finite extension of rings (again inducing a surjection on spectra). Pick a maximal ideal $\mathfrak{m}'' \subset A''$ lying over the maximal ideal of A and a discrete valuation ring A''' dominating $A''_{\mathfrak{m}''}$ (see lemma cited above). Then $A \to A'''$ is an extension of discrete valuation rings and we have $b = a^m c$ in A'''. Thus $v'''(b) \geq mv'''(a)$. Since $v''' = ev$ where e is the ramification index of A'''/A, we find that $m \leq n$ as desired. □

Lemma 1.4. *Let $(A, \mathfrak{m})$ be a Noetherian local ring of dimension 1. Let $r \geq 2$ and let $a_1, \ldots, a_r \in A$ be nonzerodivisors not all units. Then there exist*

1. *a finite ring extension $A \subset B$ with B/A annihilated by a power of $\mathfrak{m}$,*
2. *for each maximal ideal $\mathfrak{m}_j \subset B$ a nonzerodivisor $\pi_j \in B_j = B_{\mathfrak{m}_j}$, and*
3. *factorizations $a_i = u_{i,j}\pi_j^{e_{i,j}}$ in B_j with $u_{i,j} \in B_j$ units and $e_{i,j} \geq 0$.*

Proof. Since at least one a_i is not a unit we find that $\mathfrak{m}$ is not an associated prime of A. Moreover, for any $A \subset B$ as in the statement $\mathfrak{m}$ is not an associated prime of B and $\mathfrak{m}_j$ is not an associate prime of B_j. Keeping this in mind will help check the arguments below.

First, we claim that it suffices to prove the lemma for $r = 2$. We will argue this by induction on r; we suggest the reader skip the proof. Suppose we are given $A \subset B$ and π_j in $B_j = B_{\mathfrak{m}_j}$ and factorizations $a_i = u_{i,j}\pi_j^{e_{i,j}}$ for $i = 1, \ldots, r-1$ in B_j with $u_{i,j} \in B_j$ units and $e_{i,j} \geq 0$. Then by the case $r = 2$ for π_j and a_r in B_j we can find extensions $B_j \subset C_j$ and for every maximal ideal $\mathfrak{m}_{j,k}$ of C_j a nonzerodivisor $\pi_{j,k} \in C_{j,k} = (C_j)_{\mathfrak{m}_{j,k}}$ and factorizations

$$\pi_j = v_{j,k}\pi_{j,k}^{f_{j,k}} \quad \text{and} \quad a_r = w_{j,k}\pi_{j,k}^{g_{j,k}}$$

as in the lemma. There exists a unique finite extension $B \subset C$ with C/B annihilated by a power of $\mathfrak{m}$ such that $C_j \cong C_{\mathfrak{m}_j}$ for all j, see Lemma 1.1. The maximal ideals of C correspond 1-to-1 to the maximal ideals $\mathfrak{m}_{j,k}$ in the localizations and in these localizations we have

$$a_i = u_{i,j}\pi_j^{e_{i,j}} = u_{i,j}v_{j,k}^{e_{i,j}}\pi_{j,k}^{e_{i,j}f_{j,k}}$$

for $i \leq r-1$. Since a_r factors correctly too the proof of the induction step is complete.

Proof of the case $r = 2$. We will use induction on

$$\ell = \min(\mathrm{length}_A(A/a_1A),\ \mathrm{length}_A(A/a_2A)).$$

If $\ell = 0$, then either a_1 or a_2 is a unit and the lemma holds with $A = B$. Thus we may and do assume $\ell > 0$.

Suppose we have a finite extension of rings $A \subset A'$ such that A'/A is annihilated by a power of $\mathfrak{m}$ and such that $\mathfrak{m}$ is not an associated prime of A'. Let $\mathfrak{m}_1, \ldots, \mathfrak{m}_r \subset A'$ be the maximal ideals and set $A'_i = A'_{\mathfrak{m}_i}$. If we can solve the problem for a_1, a_2 in each A'_i, then we can apply Lemma 1.1 to produce a solution for a_1, a_2 in A. Choose $x \in \{a_1, a_2\}$ such that $\ell = \mathrm{length}_A(A/xA)$. By Lemma 0.2 and (0.1.1) we have $\mathrm{length}_A(A/xA) = \mathrm{length}_A(A'/xA')$. On the other hand, we have

$$\mathrm{length}_A(A'/xA') = \sum [\kappa(\mathfrak{m}_i) : \kappa(\mathfrak{m})]\mathrm{length}_{A'_i}(A'_i/xA'_i)$$

by Algebra, Lemma 02M0. Since $x \in \mathfrak{m}$ we see that each term on the right hand side is positive. We conclude that the induction hypothesis applies to a_1, a_2 in each A_i' if $r > 1$ or if $r = 1$ and $[\kappa(\mathfrak{m}_1) : \kappa(\mathfrak{m})] > 1$. We conclude that we may assume each A' as above is local with the same residue field as A.

Applying the discussion of the previous paragraph, we may replace A by the ring constructed in Lemma 1.2 for $a_1, a_2 \in A$. Then since A is local we find, after possibly switching a_1 and a_2, that $a_2 \in (a_1)$. Write $a_2 = a_1^m c$ with $m > 0$ maximal. In fact, by Lemma 1.3 we may assume m is maximal even after replacing A by any finite extension $A \subset A'$ as in the previous paragraph. If c is a unit, then we are done. If not, then we replace A by the ring constructed in Lemma 1.2 for $a_1, c \in A$. Then either (1) $c = a_1 c'$ or (2) $a_1 = c a_1'$. The first case cannot happen since it would give $a_2 = a_1^{m+1} c'$ contradicting the maximality of m. In the second case we get $a_1 = c a_1'$ and $a_2 = c^{m+1}(a_1')^m$. Then it suffices to prove the lemma for A and c, a_1'. If a_1' is a unit we're done and if not, then $\text{length}_A(A/cA) < \ell$ because cA is a strictly bigger ideal than $a_1 A$. Thus we win by induction hypothesis. $\square$